Finance Analytics & Transaction Risk Monitoring

Business Insights & Recommendations Report

Portfolio case study | Historical transaction data: January 2023–April 2026

Prepared as an analytical assessment for a hypothetical financial institution. Recommendations and targets are proposed, not implemented outcomes.

Executive summary

The analysis examines 50,000 unique transactions and 5,000 customer records to identify opportunities in transaction reliability, risk-review prioritization, customer engagement and data governance. Of all transaction attempts, 42,869 succeeded (85.74%), 5,096 failed (10.19%) and 2,035 were pending (4.07%). There were 630 fraud-flagged records (1.26%), but the dataset does not establish independently confirmed fraud. The 3,983 customers with recorded transactions represent 79.66% of the customer master.

The highest-priority opportunities are to investigate high-volume transaction failures, validate risk-score effectiveness against independent outcomes, and institutionalize transaction uniqueness and financial KPI reconciliation controls.

1. Scope and analytical definitions

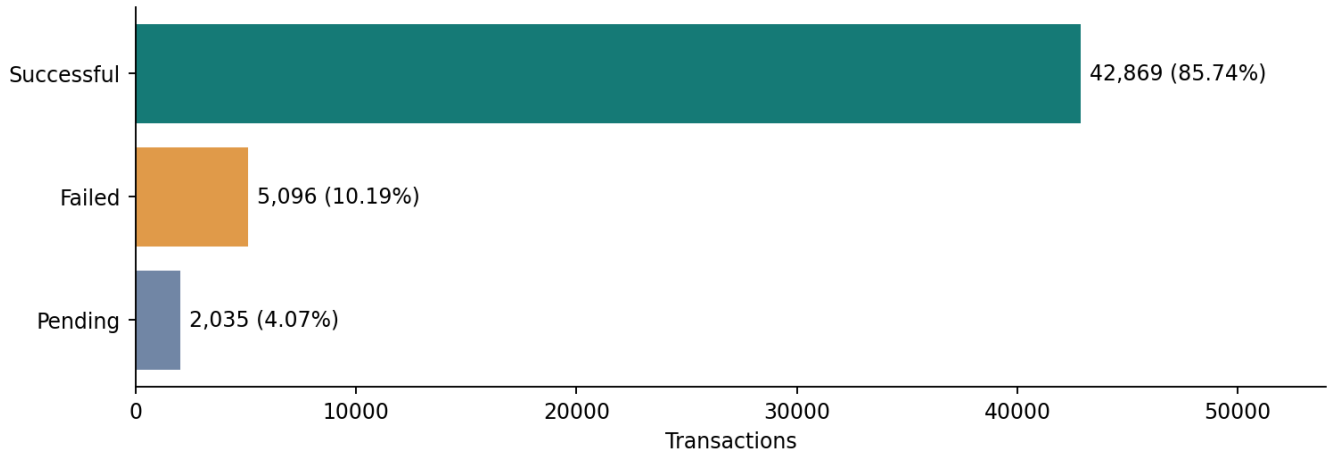
Metric	Definition / scope
Transaction attempts	50,000 distinct transaction IDs after SQL deduplication
Transaction success rate	Successful attempts / all attempts
Transaction value	Financial transaction activity; not institutional revenue
Transaction magnitude	ABS(amount); not net cash flow
Fraud flag	Dataset indicator; not necessarily confirmed fraud
Active customer	Customer with at least one recorded transaction in observed period
Fee reporting	Recorded fees plus fixed 14.52 replacement for missing values in reporting layer

2. Transaction performance and operational efficiency

Status	Count	Share

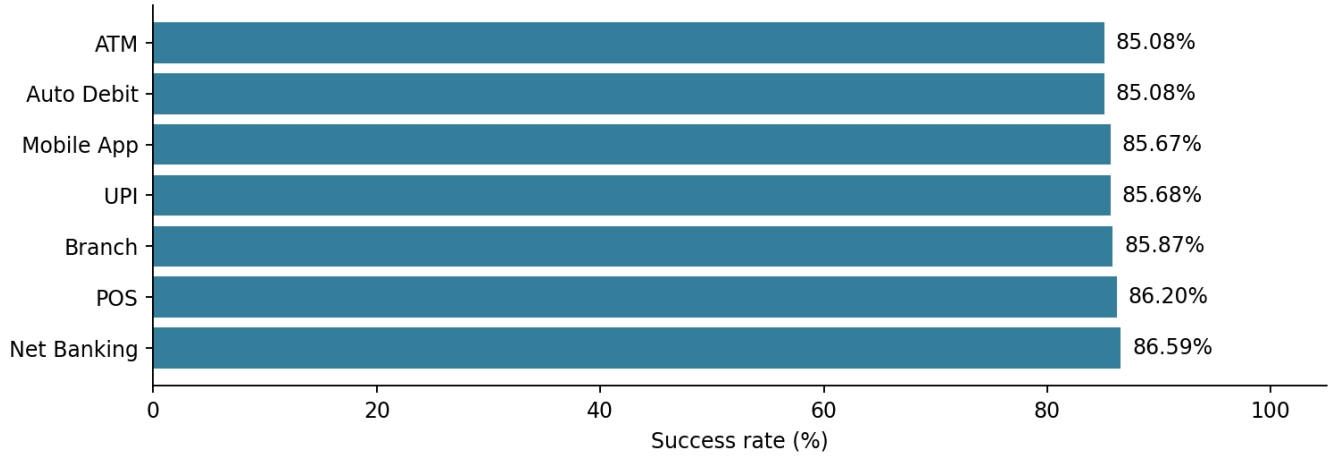
Visual analysis | Transaction reliability

Figure 1. Transaction status distribution



Source: 50,000 deduplicated transactions. Percentages use all attempts as the denominator.

Figure 2. Observed success rates by channel



The spread is small; the Python chi-square test did not establish a statistically significant channel association (p = 0.0982).

Success	42,869	85.74%
Failed	5,096	10.19%
Pending	2,035	4.07%

Channel	Attempts	Failures	Success rate
ATM	7,176	779	85.08%
Auto Debit	7,071	764	85.08%
Mobile App	7,103	703	85.67%
UPI	7,199	710	85.68%
Branch	7,228	731	85.87%
POS	7,110	727	86.20%
Net Banking	7,113	682	86.59%

Business interpretation

ATM and Auto Debit have the lowest observed success rates, although the channel spread is only about 1.51 percentage points. Loan EMI (922 failures) and Transfer (837) together account for 1,759 failures, about 34.5% of all failures. Their large volumes partly explain their failure counts; rates and volume should be evaluated together. Notable high-count channel/type combinations include Mobile App–Loan EMI (138 failures), Auto Debit–Transfer (137), and Auto Debit–Loan EMI (136).

Recommended action

Obtain failure reason codes and retry linkage; investigate high-volume channel/type combinations; compare like-for-like transaction mixes and time periods; pilot targeted operational changes. Track failure rate, success rate, pending rate, repeated attempts and customer complaints. Do not infer root causes from this dataset alone.

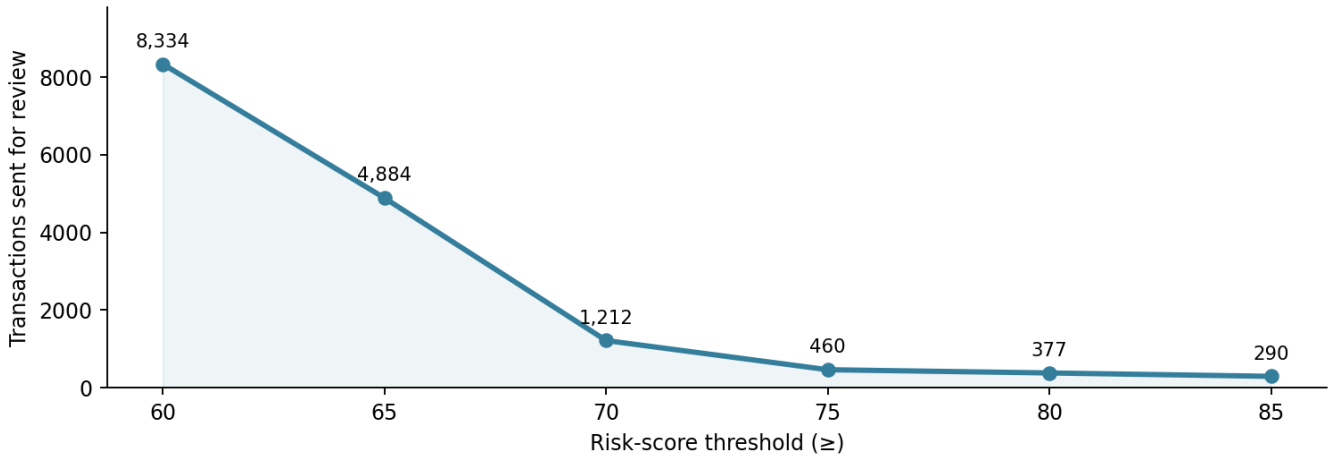
3. Risk flags and review workload

Risk measure	Observed value
Flagged transactions	630 / 50,000 (1.26%)
Flagged successful transactions	548
Average flagged risk score	82.94
Average non-flagged risk score	35.49
Flagged signed transaction value	₹5,969,443.52 (not proven loss)

Score threshold	Review queue	Existing flags included	Queue share
≥60	8,334	630 / 630	16.67%
≥65	4,884	630 / 630	9.77%
≥70	1,212	535 / 630	2.42%
≥75	460	460 / 630	0.92%
≥80	377	377 / 630	0.75%
≥85	290	290 / 630	0.58%

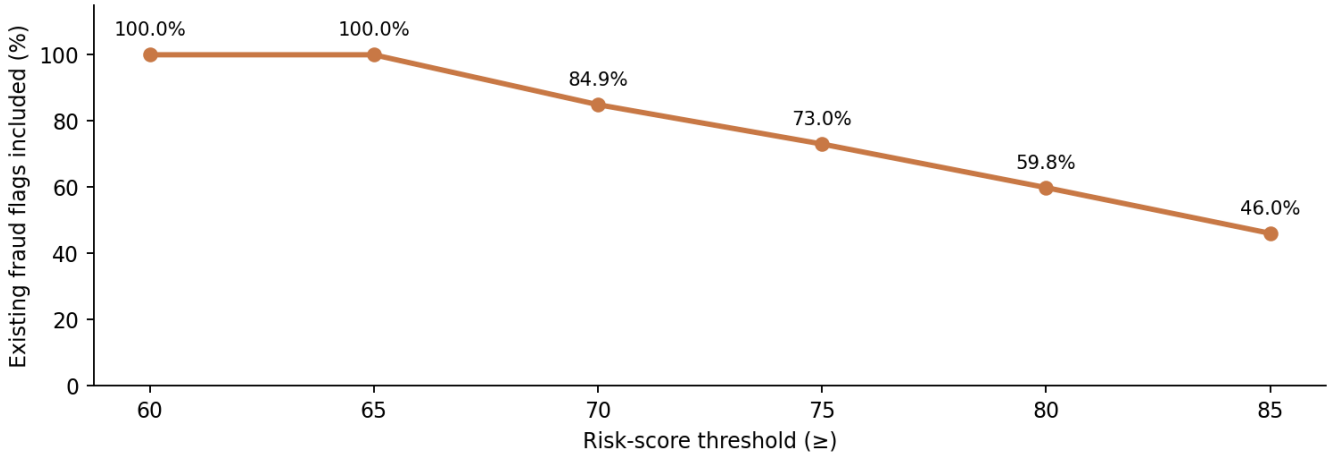
Visual analysis | Risk-review trade-offs

Figure 3. Review queue size by risk-score threshold



Higher thresholds reduce the number of transactions requiring manual review.

Figure 4. Coverage of existing fraud flags



Coverage is calculated against 630 existing dataset flags, not independently confirmed fraud. The flag may be derived from risk score.

Interpretation and limitation

All flagged records have risk scores of at least 65, while non-flagged scores extend up to 70. This unusually strong relationship suggests the fraud flag may depend on risk-score rules, creating potential label leakage. Threshold coverage against existing flags is not independent evidence of real-world fraud detection. The 548 successful flagged transactions warrant post-processing review, but their processing status does not establish a loss.

Recommended action

Validate the source and independence of labels, acquire investigator-confirmed outcomes and timestamps, and test a two-tier review queue (priority ≥75; secondary 65–74) in a controlled pilot. Monitor review volume, verified detection, false positives, turnaround time and confirmed net losses.

4. Customer activity and segmentation

Segment	Registered	With transactions	Attempts	Attempts per active customer
Retail	2,703	2,177	27,112	12.45
Premium	895	711	9,210	12.95
SME	780	616	8,002	12.99
Corporate	374	284	3,432	12.08
Wealth	248	195	2,244	11.51

Business interpretation

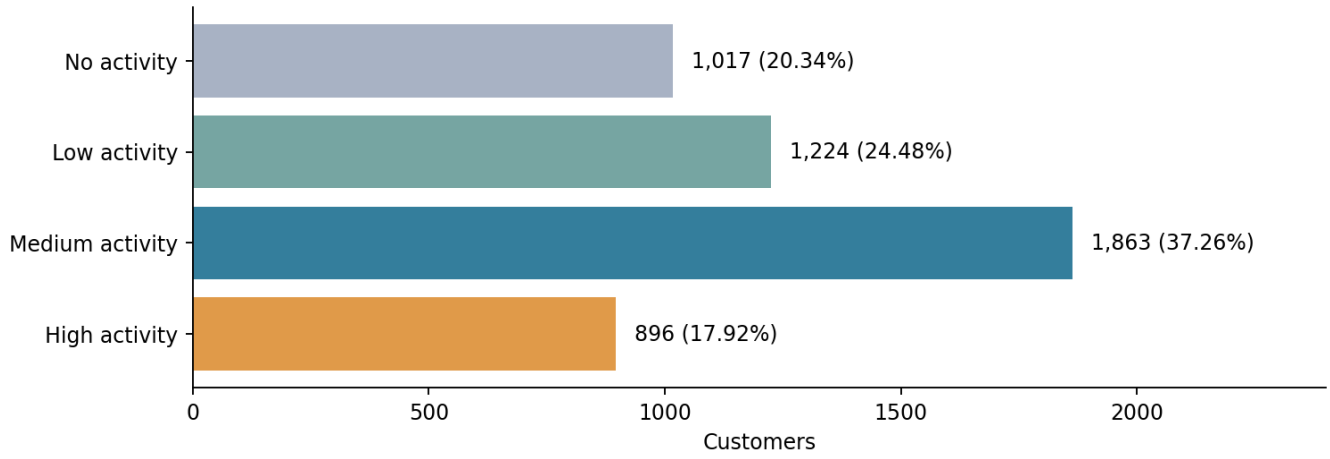
Retail accounts for the largest absolute transaction volume (54.22%). SME has the highest attempts per transaction-active customer (12.99 across the entire observation period), narrowly above Premium (12.95). Of 5,000 registered customers, 1,017 have no recorded transactions in this dataset. This does not prove they are inactive in other products. Average successful transaction amounts are relatively similar across segments; activity and profitability should not be conflated.

Recommended action

Examine SME transaction journeys, scalable Retail support, and the onboarding dates and product usage of customers with no recorded transactions. Segment customers by frequency and value, then test targeted service initiatives. Measure first-transaction conversion, cohort activity, service quality and verified retention once the requisite data is available.

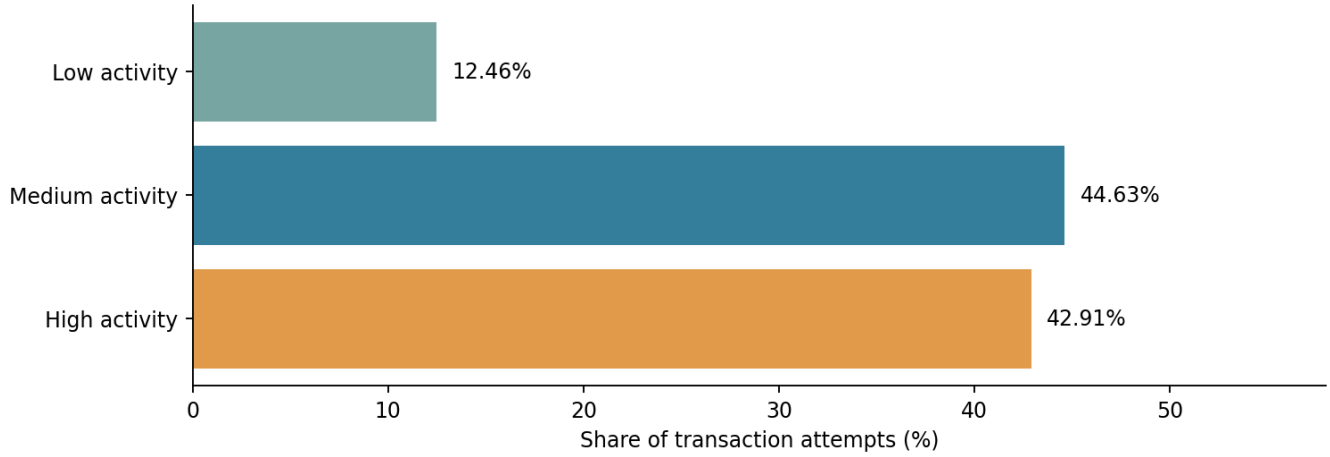
Visual analysis | Customer behavior

Figure 5. Customer frequency segments



Customer counts total 5,000, including 1,017 with no recorded transactions.

Figure 6. Transaction share by active customer cohort



The 896 high-activity customers generate 42.91% of transaction attempts; no-activity customers generate none.

5. Data quality and reporting governance

Issue	Validated finding	Control
Duplicate transaction rows	50,069 raw rows; 50,000 distinct transaction IDs	Unique-ID validation and conflicting-record review
Conflicting fee duplicate	One transaction ID had NULL versus ₹21.19	Use approved source-of-truth selection
Missing fees	23 of 50,000 validated records	Exception monitoring; recover source fee
Negative amounts	8 records	Preserve signed amounts and investigate classification
Category inconsistency	Currency capitalization and channel typo	Controlled reference lists
Customer reference integrity	All validated transaction customer IDs matched master	Automated foreign-key checks

Fee treatment

The reporting layer replaces each of 23 missing fees with ₹14.52, adding ₹333.96 of estimated fee amounts. These replacements are not verified charged fees. Report recorded fees and estimated additions separately when financial precision is required.

Reconciliation evidence

2024 validated reporting metrics: 15,029 transactions; transaction magnitude ₹135,618,901.58; average magnitude ₹9,023.81; total reporting fees ₹217,308.65; tax ₹39,142.77. SQL and the refreshed Power BI report were reconciled on these KPI definitions. Raw source records were preserved.

6. Prioritized management action plan

Priority	Initiative	Owner	Success measure
High	Investigate transaction failures	Transaction Operations / IT	Failure and success rates; reason-code mix
High	Validate risk-review framework	Fraud Risk / Compliance	Confirmed fraud outcomes; queue workload
High	Automate data controls	Data Governance / BI	Unresolved critical exceptions; reconciliation status
Medium	Profile customer activity	Customer Analytics / CRM	Activity cohorts; first-transaction conversion

7. Illustrative 90-day roadmap

Period	Deliverable
Days 1–15	Agree KPI definitions, owners and data quality checks
Days 16–30	Acquire failure diagnostics and independently confirmed fraud outcomes
Days 31–45	Select operational priorities and design risk pilot
Days 46–60	Run controlled transaction and review-workflow pilots
Days 61–75	Evaluate pilot KPIs and investigate customer cohorts
Days 76–90	Review results, cost assumptions and rollout decisions

8. Illustrative scenario and impact measurement

At a comparable 50,000 attempts, improving the success rate from 85.74% to 87.00% would correspond to approximately 630 additional successful transactions, assuming attempts previously classified as failed convert to success and pending volume remains unchanged. This is an arithmetic scenario, not a measured improvement or forecast. Realized financial impact would require failure costs, repeat-attempt data, operational expenses, and observed pilot results.

9. Data limitations and decision safeguards

The dataset lacks failure reason codes, confirmed fraud adjudications, fraud-label provenance, retry linkage, investigation costs, customer satisfaction, verified churn, product profitability and loss recoveries. Observed correlations are not causal. Transaction value is not company revenue, and flagged transaction value is not confirmed fraud loss. Management should validate data definitions and test initiatives before deploying policies.

10. Conclusion

The project demonstrates an end-to-end analytical workflow: raw data preservation, SQL validation, transaction and customer analysis, Power BI KPI reconciliation, and evidence-based recommendations. The strongest actionable next steps are transaction failure diagnosis, independent fraud-outcome validation, and repeatable data-quality governance. Benefits remain opportunities until verified by implementation and measurement.